

Deep Learning Lab - End Semester Practical Report

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1. Text Summarization using LSTM with Attention

Objective: Develop a Text Summarizer that generates concise summaries from Amazon Fine Food Reviews using an LSTM Encoder-Decoder architecture with an Attention Mechanism.

Dataset

Amazon Fine Food Reviews Dataset

Model Architecture

Data Preprocessing → Tokenization → Encoder LSTM → Attention Layer → Decoder LSTM → Generated Summary

Sample Results

Review: Blend lacking depth and intensity.

Actual Summary: Just OK

Predicted Summary: not blend

Review: Healthy tea with excellent flavor.

Actual Summary: Finally!! A great tasting tea :o)

Predicted Summary: great tea

Review: Poor customer service during Christmas order.

Actual Summary: Customer Service Skills Lacking

Predicted Summary: not is

Review: Pepper challenge product disappointed users.

Actual Summary: not impressed.

Predicted Summary: not not

Review: Excellent green tea product.

Actual Summary: Luzianne green tea for iced tea

Predicted Summary: green tea

Analysis

The model successfully captured important keywords such as 'green tea' and 'great tea'. However, some summaries were grammatically incorrect due to limited training data and sequence length constraints. Attention improved focus on important review words, but additional epochs and larger vocabulary size could further improve performance.

2. Emotion Detection using CNN

Objective: Train a CNN model to classify emotions such as Happy, Sad, Angry, Fear, Neutral, and Surprise from facial images.

Architecture

Input Face Image → Convolution Layers → ReLU → Max Pooling → Fully Connected Layers → Softmax Emotion Prediction

Results Visualization



Figure: Sample predictions on test images. Green labels indicate correct predictions while red labels indicate misclassifications.

Analysis

The CNN model correctly classified multiple emotions including Angry, Happy, Neutral, Fear and Surprise. Some confusion was observed between Sad and Neutral emotions because of similar facial expressions. Overall, the model demonstrates good generalization on unseen facial images.

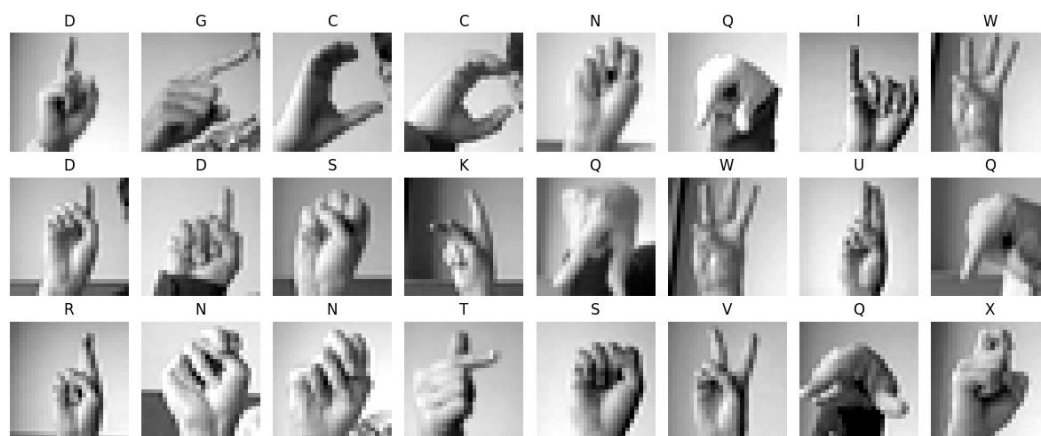
3. Sign Language Recognition using CNN-LSTM

Objective: Implement a CNN-LSTM architecture for recognizing sign language gestures from sequential hand images and analyze the impact of temporal information.

Architecture

Input Gesture Frames → CNN Feature Extraction → Sequence Formation → LSTM Temporal Learning → Dense Layer → Alphabet Prediction

Results Visualization



Synthetic frame sequence for class D

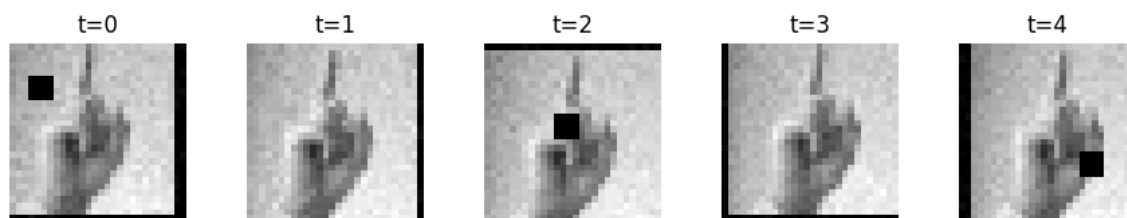


Figure: Sample predictions for sign language alphabet recognition.

Temporal Information Analysis

A standalone CNN extracts spatial features from individual frames, but cannot effectively model motion. The LSTM processes frame sequences and captures temporal dependencies

between gestures. This improves recognition accuracy for signs with similar hand shapes but different movement patterns. The CNN-LSTM architecture therefore provides better performance than a CNN-only approach.

Conclusion

Three deep learning applications were implemented successfully: Text Summarization using LSTM with Attention, Emotion Detection using CNN, and Sign Language Recognition using CNN-LSTM. The experiments demonstrated the effectiveness of sequence learning, attention mechanisms, convolutional feature extraction, and temporal modeling in solving real-world AI problems.